

Notes: Cao, Liang, Yao, and Zhang (JFE, 2026)

Liquidity Characteristics of Market Anomalies and Institutional Trading

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Goal of these notes. These notes reconstruct the full empirical logic of Cao, Liang, Yao, and Zhang (2026) and provide a coherent “model-by-model” guide to: (i) how the authors measure anomalies, liquidity, and institutional trading; (ii) the econometric decompositions they use to isolate liquidity-related trading; and (iii) how to read every table and figure as part of a single causal narrative.

1 Big Picture and Main Contribution

The paper asks a simple but deep question: *Why do institutions often appear to trade “the wrong way” on many return anomalies, and what role does liquidity play in that pattern?*

The central contribution is to show that anomaly strategies are *not liquidity-neutral*. Instead:

- **Long-horizon anomalies** (e.g., value, investment/financing, intangible, long-term profitability) have long legs that are *less liquid* and experience *less liquidity improvement* than their short legs. These strategies resemble *liquidity-provision* trades (long illiquid, short liquid).
- **Short-horizon anomalies** (e.g., earnings momentum, return momentum, short-term profitability) have the opposite liquidity pattern; the long legs are *more liquid* and experience *more liquidity improvement*. These strategies are *liquidity-demanding*.

Institutions with strong liquidity preferences (especially large institutions) respond to *exogenous liquidity shocks* (driven by corporate events, information environment changes, and tick-size reforms). Those shocks induce systematic trading on anomaly legs, generating the observed “wrong-way” trading on long-horizon anomalies and shaping anomaly magnitudes.

2 Data and Core Objects

2.1 Anomaly variables and portfolio construction

The authors study **32 anomaly variables** grouped into **7 categories**:

1. **Value:** B/M, E/P, CF/P, NPY, DUR
2. **Investment and financing:** AG, IVA, IG, PTA, CSI, NSI
3. **Intangibles:** IA, OA, AC, TC, R&D/ME
4. **Long-term profitability:** RNOA, ROA, ROE, ROIC, EQPO, CFOA
5. **Earnings momentum:** SUE, RE, FE, PEAD
6. **Return momentum:** MOM, IMOM, REV, RREV
7. **Short-term profitability:** GP, OP, F

Sorted-portfolio design (used throughout the paper). At the end of each quarter t :

1. Sort stocks into **terciles** by an anomaly signal.
2. Construct **equal-weighted** portfolios in each tercile.
3. Define the **long leg** as the tercile predicted to have high returns and the **short leg** as the tercile predicted to have low returns.
4. Hold for **8 quarters** and rebalance each holding quarter to maintain equal weights (thus returns are based on overlapping holding windows).

Delisting returns are included (Shumway-style treatment if missing), and stocks with price below \$5 at formation are excluded.

Category and aggregate portfolios. After building portfolios for each anomaly, the authors average within category (equal weights across anomalies), then average across the four long-horizon categories and three short-horizon categories to form **aggregate long-horizon** and **aggregate short-horizon** portfolios.

2.2 Institutional ownership and institutional trading

Let $\%Inst_{i,t}$ be institutional ownership (shares held by 13F institutions divided by shares outstanding), winsorized at the 99th percentile each quarter and capped at 100%. Missing ownership observations are set to 0.

Institutional trading over a window is defined as:

$$\Delta Inst_{i,t_0,t_1} \equiv \%Inst_{i,t_1} - \%Inst_{i,t_0}. \quad (1)$$

The key horizon choice is:

- Long-horizon anomalies: $\Delta Inst$ measured over **6 quarters** prior to formation.
- Short-horizon anomalies: $\Delta Inst$ measured over **2 quarters** prior to formation.

The IPO quarter is excluded when measuring trading to avoid mechanically attributing IPO-related activity to liquidity.

2.3 Liquidity: Amihud illiquidity, ranking, and liquidity changes

The primary daily illiquidity measure is Amihud (2002):

$$ILLIQ_{i,t} = \frac{1}{T} \sum_{d=1}^T \frac{|r_{i,d}|}{S_{i,d}P_{i,d}}, \quad (2)$$

computed each quarter using daily data (minimum 22 observations).

To remove long-run trends and exchange-specific reporting differences, the paper uses a **cross-sectional percentile rank** within NYSE/AMEX and within NASDAQ:

$$ILQ_{i,t} \in \{0, 1, \dots, 99\}, \quad \text{higher } ILQ \text{ means } \textit{less} \text{ liquid.}$$

Missing ILQ is assigned 99 (least liquid).

Liquidity change. Define the change in ranked illiquidity over a window:

$$\Delta ILQ_{i,t_0,t_1} \equiv ILQ_{i,t_1} - ILQ_{i,t_0}.$$

Negative ΔILQ means *liquidity improves*; positive means *liquidity deteriorates*.

The window length mirrors the institutional-trading horizon:

- Long-horizon anomalies: ΔILQ over **7 quarters** prior to formation.
- Short-horizon anomalies: ΔILQ over **3 quarters** prior to formation.

2.4 Net (long–short) objects

For any leg-level object X , define the **net** (long–short) quantity:

$$NetX \equiv X^{Long} - X^{Short}.$$

Key examples:

$$NetILQ = ILQ^{Long} - ILQ^{Short}, \quad Net\Delta ILQ = \Delta ILQ^{Long} - \Delta ILQ^{Short}, \quad Net\Delta Inst = \Delta Inst^{Long} - \Delta Inst^{Short}$$

Interpretation:

- $NetILQ > 0$: the *long* leg is less liquid than the short leg.
- $Net\Delta ILQ > 0$: the long leg experiences less liquidity improvement (or more deterioration) than the short leg.
- $Net\Delta Inst > 0$: institutions trade in the *right* direction (buy long, sell short); < 0 means they trade in the *wrong* direction.

3 Econometric “Models” Used in the Paper

This paper is empirical and its “models” are regression and decomposition designs. I list them as explicit estimating equations and constructed objects.

3.1 Model 1: Risk-adjusted anomaly returns (Fama–French alpha)

For each portfolio p , the quarterly excess return is regressed on Fama–French factors:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{p,MKT}MKT_t + \beta_{p,SMB}SMB_t + \beta_{p,HML}HML_t + \varepsilon_{p,t}. \quad (3)$$

The key quantity is α_p (reported in Table 1 and Table 11).

3.2 Model 2: Liquidity-related institutional trading decomposition

To isolate the part of institutional trading that is systematically tied to liquidity changes, the paper runs cross-sectional regressions each quarter:

$$\Delta Inst_{i,t-k,t} = a_t + b_t \Delta ILQ_{i,t-k-1,t} + \epsilon_{i,t}, \quad (4)$$

where $k = 6$ for long-horizon anomalies and $k = 2$ for short-horizon anomalies.

The regressions are run *within exchange* (NYSE/AMEX vs NASDAQ) and *within size deciles* to control for size–liquidity structure.

Constructed components.

$$\Delta Inst_{i,t-k,t}^{LIQ} \equiv \hat{a}_t + \hat{b}_t \Delta ILQ_{i,t-k-1,t} \quad (\text{fitted value}), \quad (5)$$

$$\Delta Inst_{i,t-k,t}^{NLQ} \equiv \Delta Inst_{i,t-k,t} - \Delta Inst_{i,t-k,t}^{LIQ} \quad (\text{residual}). \quad (6)$$

These are the objects in Table 4 and later used in Table 11 (Panels B and C).

3.3 Model 3: Liquidity-driving corporate events and information environment

The paper defines a set of relatively exogenous liquidity drivers:

- **Corporate events:** Financing, M&A, Dividend Change, Split, Index Change.
- **Information environment:** change in analyst coverage and change in firm-specific business news coverage.

Analyst and news changes are measured by symmetric percentage changes:

$$\% \Delta Analyst_{t-k,t} = \frac{Analyst_t - Analyst_{t-k}}{(Analyst_t + Analyst_{t-k})/2}, \quad (7)$$

$$\% \Delta News_{t-k,t} = \frac{News_t - News_{t-k}}{(News_t + News_{t-k})/2}, \quad (8)$$

with special treatment when the denominator is zero via indicator dummies *NOANALYST* and *NONNEWS*.

3.4 Model 4: Fama–MacBeth event effects on liquidity and trading

To show that these exogenous factors move liquidity and institutional trading contemporaneously, the paper runs (quarterly) cross-sectional regressions and then averages coefficients:

$$\Delta ILQ_{i,t} = \alpha_t + \beta_t Event_{i,t} + \gamma_t NoEventDummy_{i,t} + u_{i,t}, \quad (9)$$

$$\Delta Inst_{i,t} = \alpha'_t + \beta'_t Event_{i,t} + \gamma'_t NoEventDummy_{i,t} + u'_{i,t}, \quad (10)$$

where *Event* is one of the seven liquidity drivers (Table 5), and *NoEventDummy* is used when coverage-change variables are missing.

3.5 Model 5: Event-driven institutional trading decomposition

To quantify the portion of trading on anomalies attributable to liquidity-driving factors, the paper runs (each formation quarter) cross-sectional regressions:

$$\Delta Inst_{i,t-k,t} = \alpha_t + \Theta_t^\top Z_{i,t} + e_{i,t}, \quad (11)$$

where $Z_{i,t}$ collects the liquidity-driving factors (aggregated over 7 quarters for long-horizon anomalies, 3 quarters for short-horizon).

Define:

$$\Delta Inst_{i,t-k,t}^{EVT} \equiv \hat{\alpha}_t + \hat{\Theta}_t^\top Z_{i,t} \quad (\text{event-driven component}), \quad (12)$$

$$\Delta Inst_{i,t-k,t}^{NEVT} \equiv \Delta Inst_{i,t-k,t} - \Delta Inst_{i,t-k,t}^{EVT} \quad (\text{non-event residual}). \quad (13)$$

These appear in Table 7. A “filtered” version excludes events and news categories directly tied to specific anomaly definitions.

3.6 Model 6: Liquidity preference heterogeneity

Liquidity preference is proxied by institution size (the total market value of the institution’s 13F equity holdings). Institutions are sorted into terciles each quarter: *Large*, *Medium*, *Small*.

Event-response regressions (Table 8) are then run separately by group:

$$\Delta Inst_{i,t}^{(g)} = \alpha_t^{(g)} + \beta_t^{(g)} Event_{i,t} + \gamma_t^{(g)} NoEventDummy_{i,t} + u_{i,t}^{(g)}.$$

The paper also uses an alternative trading measure:

$$\Delta \#Inst_{i,t} \equiv \Delta \left(\frac{\# \text{institutional owners of } i}{\text{average } \# \text{ owners in } i\text{'s size decile}} \right),$$

to show robustness (Table 8 Panel B and Table 9 Panel B).

3.7 Model 7: Quasi-natural experiments from tick-size reforms and DiD

The minimum tick size changes in 1997 and 2001 generate liquidity shocks not driven by corporate events. Liquidity is measured by the value-weighted percentage effective spread, *Spread*.

Stocks are split into **High-shock** vs **Low-shock** based on the (above/below median) reduction in *Spread* from pre-treatment to treatment.

A difference-in-differences (DiD) estimand is:

$$DiD \equiv (\Delta Inst^{Treated,Post} - \Delta Inst^{Treated,Pre}) - (\Delta Inst^{Control,Post} - \Delta Inst^{Control,Pre}). \quad (14)$$

To support the parallel-trends assumption, the paper implements a matching-style procedure based on pre-trends in $\Delta Inst$ before assigning treated/control within anomaly legs (details in the text around Table 10 Panel C).

3.8 Model 8: Anomaly magnitude conditional on trading direction

The paper further sorts within each anomaly leg by the *direction* of institutional trading:

- Long leg: **LH** (institutions buy), **LL** (institutions sell).
- Short leg: **SL** (institutions sell), **SH** (institutions buy).

Then:

$$\text{Right portfolio} = LH - SL, \quad \text{Wrong portfolio} = LL - SH,$$

and the key comparison is:

$$(\text{Wrong} - \text{Right}) = (LL - SH) - (LH - SL),$$

reported using FF3 alphas (Table 11), for overall trading $\Delta Inst$ and for the decomposed components $\Delta Inst^{LIQ}$ and $\Delta Inst^{NLQ}$.

4 How to Read the Evidence: Table-by-Table and Figure-by-Figure

4.1 Table 1: Return-predictive horizons (classification step)

Table 1 forms the foundation: it shows quarterly post-formation returns and FF3 alphas for each anomaly category for quarters 1–8.

Key pattern.

- For **long-horizon categories**, long-short returns and alphas remain positive and statistically meaningful even at longer horizons (later holding quarters), consistent with *slow-moving correction / convergence*.
- For **short-horizon categories**, predictability is concentrated early; for example, return-momentum profitability fades quickly and can reverse, consistent with *short-run underreaction and subsequent decay*.

Why it matters. This horizon split is used repeatedly: it predicts which anomaly strategies are liquidity-provision vs liquidity-demanding and how institutions trade.

Table 1
Return-predictive horizons of market anomalies.

Panel A: Quarterly Returns (%)											
Qtr		Long Horizon					Short Horizon				
		Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.	
1	Short	2.85	2.97	2.88	2.70	2.85	2.81	2.64	2.47	2.64	
	Long	4.17	3.97	4.27	4.29	4.18	4.37	4.40	4.54	4.44	
	L-S	1.36	1.03	1.36	1.55	1.33	1.56	1.77	2.10	1.81	
		(2.34)	(4.83)	(3.99)	(5.67)	(4.65)	(4.95)	(4.24)	(5.04)	(5.47)	
	t-stat	2.77	2.97	2.93	2.70	2.84	3.08	2.98	2.70	2.92	
2	Short	4.12	3.81	4.13	4.18	4.06	4.00	3.81	4.07	3.96	
	Long	1.46	0.88	1.17	1.47	1.25	0.93	0.88	1.44	1.08	
	L-S	2.65	(4.58)	(3.81)	(5.58)	(4.96)	(4.13)	(2.51)	(3.77)	(4.11)	
		2.67	2.88	2.88	2.65	2.77	3.34	3.31	2.92	3.19	
	t-stat	4.01	3.72	3.95	4.02	3.93	3.66	3.25	3.51	3.48	
3	Short	1.42	0.88	1.04	1.33	1.16	0.35	-0.03	0.63	0.32	
	Long	2.85	(5.01)	(3.87)	(5.03)	(5.13)	(1.82)	(-0.08)	(1.69)	(1.33)	
	L-S	2.77	3.00	2.98	2.70	2.86	3.66	3.57	3.24	3.49	
		4.01	3.72	3.99	4.04	3.94	3.45	3.14	3.41	3.33	
	t-stat	1.37	0.75	0.96	1.33	1.10	-0.17	-0.44	0.28	-0.11	
4	Short	(2.90)	(4.92)	(3.42)	(5.22)	(5.55)	(-0.91)	(-1.43)	(0.79)	(-0.51)	
	Long	2.83	3.05	2.98	2.78	2.91	3.51	3.75	3.15	3.47	
	L-S	3.90	3.68	3.97	3.99	3.88	3.61	3.03	3.61	3.42	
		1.20	0.67	0.91	1.19	0.99	0.11	-0.77	0.53	-0.04	
	t-stat	(2.58)	(4.21)	(3.37)	(4.95)	(5.18)	(0.58)	(-2.45)	(1.62)	(-0.20)	
5	Short	3.07	3.26	3.18	3.03	3.13	3.60	3.75	3.41	3.59	
	Long	3.99	3.77	4.11	4.09	3.99	3.76	3.29	3.61	3.56	
	L-S	1.05	0.54	0.86	1.04	0.87	0.15	-0.55	0.25	-0.05	
		(2.21)	(3.02)	(3.10)	(4.54)	(4.49)	(0.76)	(-1.85)	(0.70)	(-0.21)	
	t-stat	3.44	3.59	3.46	3.30	3.45	3.92	4.04	3.78	3.92	
6	Short	4.19	3.99	4.43	4.33	4.23	4.02	3.58	3.73	3.78	
	Long	0.88	0.43	0.86	0.99	0.79	0.08	-0.52	0.02	-0.14	
	L-S	(1.76)	(2.57)	(2.98)	(4.18)	(3.88)	(0.38)	(-1.55)	(0.05)	(-0.56)	
		3.54	3.68	3.58	3.40	3.55	3.88	4.12	3.78	3.93	
	t-stat	4.11	3.93	4.37	4.32	4.18	3.99	3.51	3.83	3.78	
7	Short	0.69	0.69	0.69	0.87	0.63	0.09	-0.67	0.11	-0.15	
	Long	(1.42)	(1.46)	(2.34)	(3.72)	(3.02)	(0.36)	(-1.73)	(0.31)	(-0.54)	
Panel B: Three-factor Alpha (%)											
1	Short	-0.95	-0.81	-0.78	-0.99	-0.88	-1.03	-1.26	-1.42	-1.23	
	Long	(-4.82)	(-4.81)	(-4.97)	(-5.11)	(-5.69)	(-5.49)	(-5.05)	(-6.50)	(-6.47)	
	L-S	0.53	0.92	0.68	0.99	0.61	0.50	0.91	0.82	0.82	
		(2.48)	(2.44)	(2.68)	(3.70)	(3.40)	(3.28)	(3.06)	(4.49)	(3.89)	
	t-stat	1.84	1.13	1.29	1.67	1.39	1.69	2.04	2.43	2.06	
2	Short	(5.08)	(6.28)	(5.60)	(5.79)	(7.11)	(5.59)	(4.60)	(6.48)	(6.31)	
	Long	-0.93	-0.73	-0.67	-0.92	-0.81	-0.69	-0.87	-1.11	-0.89	
	L-S	(-4.98)	(-3.99)	(-4.42)	(-5.42)	(-5.42)	(-4.25)	(-3.77)	(-5.37)	(-5.08)	
		0.56	0.23	0.45	0.64	0.47	0.40	0.31	0.60	0.44	
	t-stat	(2.95)	(1.91)	(2.19)	(3.24)	(2.88)	(2.19)	(1.44)	(2.98)	(2.38)	
3	Short	1.48	0.97	1.13	1.55	1.28	1.09	1.18	1.72	1.33	
	Long	(5.64)	(5.94)	(4.31)	(5.98)	(7.33)	(4.96)	(3.32)	(5.26)	(5.25)	
	L-S	-0.91	-0.74	-0.65	-0.90	-0.80	-0.37	-0.44	-0.81	-0.54	
		(-4.84)	(-4.07)	(-4.22)	(-5.24)	(-5.23)	(-2.25)	(-1.78)	(-3.80)	(-3.04)	
	t-stat	0.47	0.20	0.35	0.56	0.40	0.15	-0.18	0.16	0.04	
4	Short	(2.57)	(1.56)	(1.72)	(2.72)	(4.07)	(0.88)	(-0.98)	(0.71)	(0.25)	
	Long	1.37	0.94	1.00	1.47	1.20	0.51	0.26	0.96	0.58	
	L-S	(5.42)	(6.29)	(3.90)	(5.46)	(6.91)	(2.97)	(0.81)	(2.80)	(2.73)	
		-0.84	-0.63	-0.58	-0.93	-0.75	-0.26	-0.55	-0.29	-0.29	
	t-stat	(-4.01)	(-3.24)	(-3.26)	(-5.19)	(-4.31)	(-0.39)	(-1.09)	(-2.49)	(-1.50)	
5	Short	0.44	0.17	0.36	0.60	0.39	-0.09	-0.28	0.03	-0.11	
	Long	(2.21)	(1.12)	(3.61)	(2.48)	(2.23)	(-0.53)	(-1.52)	(0.15)	(-0.60)	
	L-S	1.28	0.80	0.94	1.53	1.14	-0.02	-0.01	0.58	0.18	
		(5.08)	(5.76)	(3.69)	(6.09)	(7.18)	(-0.11)	(-0.05)	(1.90)	(0.89)	
	t-stat	-0.77	-0.57	-0.57	-0.86	-0.69	-0.20	-0.05	-0.65	-0.30	
6	Short	(-3.38)	(-2.75)	(-2.87)	(-4.52)	(-3.61)	(-0.95)	(-0.19)	(-2.97)	(-1.88)	
	Long	0.35	0.14	0.34	0.56	0.35	0.08	-0.40	0.23	-0.03	
	L-S	(1.64)	(0.89)	(1.55)	(2.43)	(1.90)	(0.45)	(-2.24)	(1.22)	(-0.20)	
		1.12	0.72	0.91	1.41	1.04	0.27	-0.35	0.88	0.27	
	t-stat	(4.02)	(5.16)	(3.72)	(5.85)	(6.40)	(1.73)	(-1.14)	(3.58)	(1.47)	
7	Short	-0.56	-0.39	-0.39	-0.62	-0.49	-0.12	-0.01	-0.41	-0.18	
	Long	(-2.14)	(-1.79)	(-1.76)	(-3.14)	(-2.33)	(-0.59)	(-0.05)	(-1.68)	(-0.78)	
	L-S	0.41	0.22	0.47	0.65	0.44	0.18	-0.24	0.22	0.05	
		(1.99)	(1.37)	(2.10)	(2.78)	(2.42)	(0.95)	(-1.18)	(1.10)	(0.30)	
	t-stat	0.97	0.61	0.86	1.26	0.93	0.31	-0.23	0.63	0.24	
8	Short	(3.09)	(4.34)	(3.37)	(5.75)	(5.60)	(1.67)	(-0.67)	(2.34)	(1.10)	
	Long	-0.35	-0.19	-0.23	-0.44	-0.31	0.07	0.21	-0.17	0.04	
	L-S	(-1.12)	(-0.81)	(-0.93)	(-1.99)	(-1.26)	(0.28)	(0.67)	(-0.62)	(0.14)	
		0.53	0.34	0.64	0.75	0.57	0.35	-0.10	0.22	0.16	
	t-stat	(2.46)	(1.93)	(2.61)	(2.99)	(2.91)	(1.65)	(-0.58)	(1.04)	(0.85)	
9	Short	0.89	0.53	0.87	1.20	0.87	0.28	-0.31	0.39	0.12	
	Long	(2.46)	(3.82)	(3.25)	(4.87)	(4.64)	(1.45)	(-1.03)	(1.41)	(0.57)	
	L-S	-0.24	-0.07	-0.09	-0.37	-0.19	0.04	0.40	-0.13	0.10	
		(-0.76)	(-0.29)	(-0.36)	(-1.60)	(-0.79)	(0.16)	(1.18)	(-0.48)	(0.37)	
	t-stat	0.49	0.29	0.60	0.79	0.54	0.35	-0.20	0.35	0.17	
10	Short	(2.33)	(1.61)	(2.42)	(3.34)	(2.78)	(1.56)	(-0.89)	(1.56)	(0.80)	
	Long	0.73	0.36	0.69	1.16	0.73	0.31	-0.60	0.49	0.07	
	L-S	(2.04)	(2.47)	(2.60)	(4.73)	(3.85)	(1.37)	(-1.48)	(1.59)	(0.25)	

Table 2
Liquidity characteristics of anomaly portfolios.

Panel A: Level of Illiquidity (ILQ)											
	Long Horizon					Short Horizon					
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.		
Short	37.03	39.78	34.92	42.28	38.50	40.89	42.21	45.49	42.86		
Long	44.73	42.27	44.18	36.82	42.00	36.05	38.33	34.57	36.32		
L-S	7.70	2.48	9.26	-5.46	3.50	-4.84	-3.88	-10.92	-6.55		
t-stat	(7.05)	(5.29)	(23.84)	(-7.73)	(6.72)	(-15.91)	(-5.66)	(-20.83)	(-17.65)		
Panel B: Change of Illiquidity (ΔILQ)											
	Long Horizon					Short Horizon					
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.		
Short	-10.86	-9.52	-8.59	-7.36	-9.08	-0.53	0.48	-2.91	-0.99		
Long	-3.16	-4.74	-5.81	-7.68	-5.35	-4.60	-6.32	-4.56	-5.16		
L-S	7.71	4.78	2.78	-0.32	3.74	-4.07	-6.80	-1.65	-4.18		
t-stat	(10.30)	(11.22)	(5.43)	(-0.50)	(8.66)	(-15.43)	(-22.94)	(-3.19)	(-12.54)		

4.2 Table 2 and Fig. 1 (Panels A–B): Liquidity characteristics around formation

Table 2 quantifies liquidity levels (ILQ) at formation and prior liquidity changes (ΔILQ).

Table 2 Panel A (levels).

- Long-horizon anomalies (on average) have $NetILQ > 0$: the long legs are *less liquid* than the short legs.
- Short-horizon anomalies have $NetILQ < 0$: the long legs are *more liquid*.

Table 2 Panel B (pre-formation liquidity change).

- Long-horizon anomalies have $Net\Delta ILQ > 0$: the short leg experiences *more liquidity improvement* than the long leg before formation.

- Short-horizon anomalies have $Net\Delta ILQ < 0$: the long leg experiences *more liquidity improvement*.

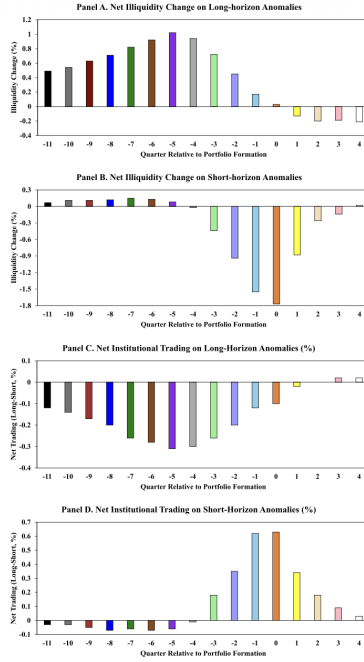


Fig. 1. Quarterly institutional trading and liquidity change around portfolio formation.

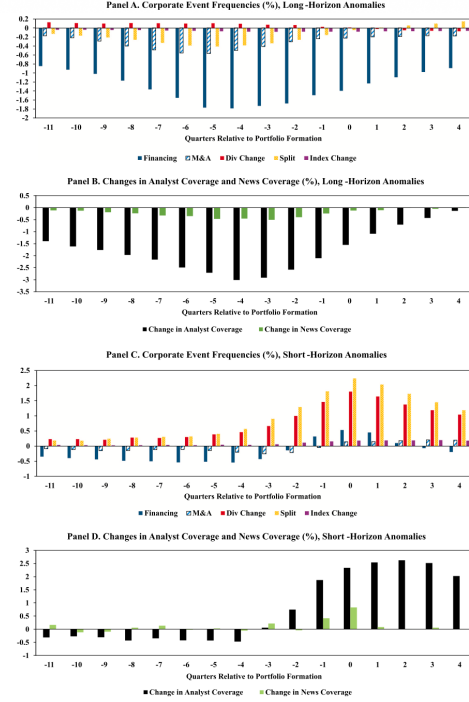


Fig. 2. Frequencies of corporate events, change in analyst coverage and change in news coverage around portfolio formation.

Interpretation. Long-horizon anomaly strategies are like on/off-the-run convergence trades: being *long illiquid and short liquid* gives the strategy a *liquidity-provision* footprint. Short-horizon strategies are *liquidity-demanding* (long liquid, short illiquid).

Fig. 1 Panels A–B (dynamics). Fig. 1 extends Table 2 by plotting net ΔILQ over a 16-quarter window around formation. For long-horizon portfolios, net ΔILQ is generally positive pre-formation (short legs improving more), then flips post-formation. For short-horizon portfolios, net ΔILQ becomes sharply negative in the few quarters right around formation (long legs improving more).

4.3 Table 3 and Fig. 1 (Panels C–D): Institutional trading on anomalies

Table 3 measures $Net\Delta Inst$ pre-formation.

Core result.

- For **long-horizon anomalies**, $Net\Delta Inst < 0$: institutions increase ownership *more* in the *short* legs than in the long legs, i.e., trading in the *wrong direction*.
- For **short-horizon anomalies**, $Net\Delta Inst > 0$: institutions trade in the *right direction*.

Fig. 1 Panels C–D. The time series of net $\Delta Inst$ mirrors the net ΔILQ pattern, especially for long-horizon anomalies: institutions buy more precisely where liquidity improves more.

Table 3

Institutional trading on market anomalies.

	Long Horizon					Short Horizon			
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Short	5.60	4.91	4.52	4.10	4.78	0.71	0.37	1.20	0.76
Long	2.62	3.47	3.84	4.46	3.60	1.98	2.25	1.76	2.00
L-S	-2.98	-1.44	-0.68	0.36	-1.18	1.26	1.88	0.56	1.24
t-stat	(-9.23)	(-9.87)	(-3.80)	(1.07)	(-7.23)	(7.49)	(9.78)	(2.63)	(6.86)

This table reports institutional trading (in percentage points, %) on anomaly portfolios prior to portfolio formation. Institutional trading ($\Delta Inst$) is the change in the percentage of shares held by institutions, measured over six (two) quarters for long-horizon (short-horizon) anomalies. In each quarter, we calculate the average $\Delta Inst$ for the long and short legs, then calculate the long-short difference (L-S) for each anomaly portfolio. Subsequently, we average them across anomalies within the same category and further average them over time. LH Avg. and SH Avg. are the averages across the four long-horizon and three short-horizon categories, respectively. t-statistics reported in parentheses are computed using Newey-West heteroskedasticity and autocorrelation consistent standard errors.

4.4 Table 4: Liquidity-related vs non-liquidity trading

Table 4 applies the decomposition in Eq. (4).

Main takeaway. For long-horizon anomalies, most of the negative $Net\Delta Inst$ is explained by the *liquidity-related* component $\Delta Inst^{LIQ}$; the residual $\Delta Inst^{NLQ}$ is small on average. For short-horizon anomalies, both components can be positive, but the liquidity-related component is still meaningfully aligned with the liquidity patterns.

Interpretation. The “wrong-way” institutional trading on long-horizon anomalies is largely a *liquidity response*, not simply disagreement about fundamentals.

Table 5

Effects of Corporate Events, Analyst Coverage Change and News Coverage Change on Liquidity and Institutional Trading.

Panel A: Quarterly Frequencies (%) of Corporate Events

Event	Data	Frequency	Event	Data	Frequency
Financing	1980-2021	2.44	% Δ Analyst	1985-2021	70.69
M&A	1980-2021	1.42	% Δ News	2001-2021	59.41
Dividend Change	1980-2021	2.95			
Split	1980-2021	1.50			
Index Change	1980-2021	0.15			

Panel B: Impact on Liquidity Change and Institutional Trading

Event Type	Dep. Var.: ΔILQ (%)			Dep. Var.: $\Delta Inst$ (%)		
	Intercept	Event	No-event Dummy	Intercept	Event	No-event Dummy
Financing	-0.73	-3.74		0.55	6.81	
r-stat	(-4.48)	(-21.88)		(6.94)	(13.68)	
M&A	-0.82	-0.90		0.70	0.86	
r-stat	(-5.08)	(-5.24)		(8.72)	(3.29)	
Dividend Change	-0.83	-0.92		0.72	0.37	
r-stat	(-5.16)	(-8.66)		(8.87)	(5.70)	
Split	-0.81	-0.92		0.71	-2.11	
r-stat	(-5.12)	(-4.43)		(8.73)	(-1.85)	
Index Change	-0.83	-0.84		0.71	2.81	
r-stat	(-5.17)	(-3.32)		(8.80)	(4.51)	
% Δ Analyst	-0.27	-3.07	-1.37	0.64	1.17	0.23
r-stat	(-4.44)	(-12.56)	(-8.43)	(6.68)	(12.19)	(1.58)
% Δ News	-0.40	-2.31	-0.62	0.70	1.04	0.31
r-stat	(-5.38)	(-9.66)	(-4.00)	(5.16)	(7.06)	(2.54)

4.5 Table 5: Exogenous factors move liquidity and institutional trading

Table 5 documents two facts:

1. Corporate events, analyst coverage changes, and news coverage changes are empirically associated with *liquidity improvements* (negative ΔILQ).
2. The same factors are associated with *higher institutional trading* (positive $\Delta Inst$).

This is crucial for causality: it provides plausible *exogenous drivers of liquidity* that can explain why certain anomaly legs systematically become more liquid.

Table 4

Liquidity-related institutional trading on market anomalies.

	Long horizon					Short horizon			
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Liquidity-Related Institutional Trading $\Delta Inst^{LIQ}$ (%)									
Short	4.85	4.33	4.26	3.84	4.32	0.92	0.70	1.24	0.95
Long	2.60	3.17	3.40	3.89	3.27	1.52	1.68	1.39	1.53
L-S	-2.25	-1.16	-0.86	0.06	-1.05	0.60	0.98	0.14	0.58
t-stat	(-9.96)	(-12.34)	(-5.21)	(0.30)	(-8.96)	(9.64)	(10.95)	(1.61)	(8.80)
Non-Liquidity Institutional Trading $\Delta Inst^{NLQ}$ (%)									
Short	0.76	0.58	0.25	0.26	0.46	-0.21	-0.33	-0.04	-0.19
Long	0.02	0.30	0.43	0.57	0.33	0.45	0.57	0.37	0.47
L-S	-0.74	-0.28	0.18	0.31	-0.13	0.66	0.90	0.42	0.66
t-stat	(-4.34)	(-3.04)	(1.59)	(1.63)	(-1.31)	(4.69)	(5.64)	(2.98)	(4.61)

Table 6

Exogenous liquidity-driving factors and market anomalies.

	Long Horizon					Short Horizon			
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Panel A. Financing, 1980-2021									
Short	25.46	22.43	19.05	23.83	22.69	6.85	6.12	9.01	7.32
Long	9.48	11.60	12.57	11.46	11.28	8.52	9.10	6.51	8.04
L-S	-15.98	-10.84	-6.47	-12.37	-11.42	1.67	2.98	-2.49	0.72
t-stat	(-6.65)	(-11.39)	(-6.98)	(-5.29)	(-8.91)	(4.32)	(7.92)	(-2.21)	(1.54)
Panel B. M&A, 1980-2021									
Short	10.49	12.56	11.13	10.58	11.19	4.51	4.28	4.61	4.47
Long	9.10	8.12	8.08	8.26	8.39	4.36	4.64	4.05	4.35
L-S	-1.40	-4.45	-3.05	-2.32	-2.80	-0.15	0.57	-0.56	-0.12
t-stat	(-3.50)	(-8.47)	(-7.25)	(-7.13)	(-7.46)	(-1.31)	(2.48)	(-3.61)	(-1.05)
Panel C. Dividend Change, 1980-2021									
Short	3.06	6.36	6.04	0.86	4.08	-0.44	0.06	-0.86	-0.41
Long	4.56	1.87	3.25	8.01	4.42	3.90	3.27	4.58	3.92
L-S	1.50	-4.49	-2.79	7.15	0.34	4.34	3.20	5.44	4.33
t-stat	(1.56)	(-5.40)	(-5.44)	(7.30)	(0.89)	(8.25)	(8.10)	(9.23)	(8.80)
Panel D. Split, 1980-2021									
Short	12.38	13.22	13.49	5.63	11.18	2.20	2.14	1.51	1.95
Long	6.52	6.65	7.29	15.37	8.96	7.48	7.01	7.67	7.38
L-S	-5.86	-6.57	-6.20	9.74	-2.22	5.27	4.87	6.15	5.43
t-stat	(-3.28)	(-6.71)	(-5.84)	(10.07)	(-2.88)	(6.91)	(7.03)	(8.77)	(7.65)
Panel E. Index Change, 1980-2021									
Short	1.17	1.26	1.45	0.45	1.08	0.13	0.17	0.06	0.12
Long	0.26	0.28	0.21	1.31	0.52	0.59	0.46	0.70	0.58
L-S	-0.91	-0.98	-1.24	0.86	-0.57	0.46	0.29	0.63	0.46
t-stat	(-5.60)	(-10.34)	(-7.79)	(7.50)	(-6.70)	(9.45)	(5.46)	(10.70)	(9.70)
Panel F. Change in Analyst Coverage (% $\Delta Analyst$), 1985-2021									
Short	30.71	31.73	26.51	18.97	26.98	6.58	6.06	9.77	7.47
Long	6.06	7.25	11.32	22.36	11.75	11.96	11.54	12.72	12.07
L-S	-24.65	-24.48	-15.19	3.39	-15.23	5.37	5.48	2.96	4.60
t-stat	(-11.16)	(-10.12)	(-8.20)	(1.42)	(-11.57)	(6.36)	(6.59)	(1.60)	(4.34)
Panel G. Change in News Coverage (% $\Delta News$), 2001-2021									
Short	14.10	14.99	13.80	14.81	14.43	4.69	3.42	6.89	5.00
Long	11.65	11.90	12.19	12.62	12.09	6.38	6.39	5.55	6.11
L-S	-2.46	-3.09	-1.61	-2.19	-2.34	1.68	2.97	-1.34	1.11
t-stat	(-1.31)	(-6.08)	(-2.14)	(-1.47)	(-2.09)	(5.78)	(13.27)	(-1.43)	(2.93)
Panel H. Change in Filtered News Coverage (Filtered % $\Delta News$), 2001-2021									
Short	14.03	15.25	13.80	14.41	14.37	4.32	3.42	6.56	4.77
Long	11.50	11.97	12.19	12.58	12.06	6.43	6.39	5.46	6.09
L-S	-2.53	-3.28	-1.61	-1.82	-2.31	2.11	2.97	-1.10	1.32
t-stat	(-1.37)	(-6.36)	(-2.14)	(-1.14)	(-2.05)	(5.95)	(13.27)	(-1.16)	(3.35)

Table 7
Event-driven institutional trading on anomalies.

Panel A: Based on Unfiltered Liquidity-driving Factors									
	Long Horizon					Short Horizon			
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Event-driven Institutional Trading, $\Delta Inst^{EVT}$ (%)									
Short	3.80	3.72	3.41	3.29	3.56	0.89	0.85	1.00	0.91
Long	2.24	2.41	2.55	2.93	2.53	1.07	1.07	1.00	1.05
L-S	-1.56	-1.31	-0.86	-0.36	-1.02	0.18	0.23	0.00	0.14
t-stat	(-9.78)	(-9.36)	(-6.77)	(-2.19)	(-9.80)	(5.85)	(7.37)	(0.02)	(3.62)
Non-event Institutional Trading, $\Delta Inst^{NEVT}$ (%)									
Short	1.81	1.19	1.12	0.81	1.23	-0.17	-0.48	0.20	-0.15
Long	0.37	1.06	1.28	1.54	1.06	0.91	1.18	0.77	0.95
L-S	-1.44	-0.14	0.17	0.73	-0.17	1.08	1.65	0.57	1.10
t-stat	(-6.38)	(-1.12)	(1.47)	(3.37)	(-1.53)	(7.74)	(9.60)	(3.70)	(7.49)

Panel B: Based on Filtered Liquidity-driving Factors									
	Long Horizon					Short Horizon			
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Event-driven Institutional Trading, $\Delta Inst^{EVT}$ (%)									
Short	3.80	3.62	3.41	3.29	3.53	0.85	0.85	1.00	0.90
Long	2.25	2.49	2.55	2.93	2.55	1.02	1.07	1.00	1.03
L-S	-1.55	-1.13	-0.86	-0.36	-0.97	0.17	0.23	0.00	0.13
t-stat	(-9.70)	(-9.16)	(-6.77)	(-2.17)	(-9.75)	(5.81)	(7.37)	(0.07)	(3.59)
Non-event Institutional Trading, $\Delta Inst^{NEVT}$ (%)									
Short	1.82	1.30	1.12	0.81	1.26	-0.14	-0.48	0.20	-0.14
Long	0.37	0.98	1.28	1.53	1.04	0.96	1.18	0.76	0.97
L-S	-1.45	-0.32	0.17	0.72	-0.22	1.10	1.65	0.56	1.10
t-stat	(-6.43)	(-2.64)	(1.47)	(3.33)	(-1.97)	(7.67)	(9.60)	(3.67)	(7.45)

4.6 Table 6 and Fig. 2: Anomalies load on liquidity-driving factors

Table 6 compares the long vs short legs in terms of exposure to liquidity-driving events and information-environment changes.

Long-horizon anomalies. Long legs have *fewer* liquidity-driving corporate events and *smaller* improvements in analyst and news coverage than short legs. This aligns one-for-one with their worse liquidity outcomes.

Short-horizon anomalies. Long legs tend to have *more* dividend changes/splits/index changes and *larger* increases in analyst/news coverage close to formation, again aligning with their liquidity improvements.

Fig. 2 (dynamics). Fig. 2 shows these exposure patterns quarter-by-quarter around formation, reinforcing the idea that liquidity patterns are *driven by event timing* rather than coincidental correlations.

4.7 Table 7: Event-driven institutional trading on anomalies

Table 7 implements the event-driven decomposition in Eq. (11). **Long-horizon anomalies.** The *event-driven* component $\Delta Inst^{EVT}$ is strongly negative and explains the bulk of wrong-way trading; the residual component is small.

Short-horizon anomalies. Event-driven trading explains little; the residual component dominates.

Filtered events check. Excluding event/news categories directly tied to certain anomaly definitions changes results only modestly, supporting the view that liquidity events matter beyond purely mechanical links.

Table 8

Liquidity preferences and institutional trading response to events

Panel A: Dependent Variable ΔInst									
Event Type	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Intercept	Event	No-event Dummy	Intercept	Event	No-event Dummy	Intercept	Event	No-event Dummy
	Large Institutions			Medium Institutions			Small Institutions		
Financing	0.32	4.51		0.03	0.32		0.02	0.13	
r-stat	(4.88)	(9.20)		(3.14)	(5.31)		(3.39)	(4.59)	
M&A	0.43	0.35		0.04	0.15		0.02	-0.02	
r-stat	(6.19)	(4.06)		(3.52)	(2.63)		(3.94)	(-0.64)	
Dividend Change	0.43	0.48		0.04	-0.03		0.02	-0.04	
r-stat	(6.32)	(7.69)		(3.80)	(-0.78)		(3.91)	(-1.99)	
Split	0.43	-2.89		0.04	-0.89		0.02	-0.51	
r-stat	(6.25)	(-1.99)		(3.51)	(-3.41)		(3.83)	(-3.16)	
Index Change	0.43	3.66		0.04	0.02		0.02	-0.01	
r-stat	(6.22)	(4.12)		(3.68)	(0.23)		(3.89)	(-0.36)	
%Analyst	0.51	0.72	-0.26	0.03	0.04	0.04	0.02	0.01	0.02
r-stat	(5.95)	(8.25)	(-3.90)	(2.32)	(1.94)	(2.29)	(2.62)	(1.03)	(1.97)
%News	0.53	0.71	0.05	0.03	0.10	0.04	0.02	0.05	0.02
r-stat	(4.12)	(3.33)	(0.77)	(1.70)	(6.06)	(2.73)	(2.07)	(2.92)	(2.34)

Panel B: Dependent Variable ΔInst									
Event Type	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Intercept	Event	No-event Dummy	Intercept	Event	No-event Dummy	Intercept	Event	No-event Dummy
	Large Institutions			Medium Institutions			Small Institutions		
Financing	2.26	18.33		0.40	4.11		0.24	2.18	
r-stat	(6.83)	(7.94)		(3.84)	(6.17)		(4.27)	(3.74)	
M&A	2.60	1.35		0.47	0.95		0.28	0.20	
r-stat	(8.11)	(3.35)		(4.58)	(1.80)		(5.10)	(2.70)	
Dividend Change	2.62	2.36		0.48	0.59		0.28	0.26	
r-stat	(8.25)	(9.56)		(4.70)	(6.18)		(5.17)	(4.03)	
Split	2.58	-3.40		0.46	0.11		0.27	0.26	
r-stat	(8.31)	(-1.34)		(4.64)	(0.22)		(5.16)	(0.52)	
Index Change	2.62	16.90		0.48	3.80		0.29	1.35	
r-stat	(8.23)	(6.44)		(4.70)	(5.54)		(5.16)	(5.47)	
%Analyst	2.45	3.08	-0.53	0.32	0.41	-0.10	0.22	0.26	-0.04
r-stat	(7.40)	(16.09)	(-3.02)	(5.89)	(8.05)	(-2.38)	(5.12)	(6.71)	(-1.55)
%News	2.02	2.03	0.06	0.38	0.38	0.04	0.19	0.25	0.07
r-stat	(6.84)	(10.43)	(0.34)	(5.06)	(11.30)	(1.08)	(4.18)	(7.82)	(1.94)

Table 9

Liquidity preferences and trading on anomalies.

Panel A: Institutional Trading based on ΔInst									
Long Horizon					Short Horizon				
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Large Institutions									
Short	5.92	5.30	4.98	4.45	5.16	0.83	0.44	1.36	0.88
Long	2.85	3.72	3.99	4.68	3.81	1.89	2.99	1.72	1.90
L-S	-3.07	-1.57	-0.99	0.24	-1.35	1.05	1.65	0.36	1.02
t-stat	(-8.23)	(-11.73)	(-5.56)	(0.69)	(-6.86)	(8.03)	(9.50)	(1.95)	(7.20)
Medium Institutions									
Short	0.15	0.04	-0.02	-0.01	0.04	-0.10	-0.08	-0.04	-0.07
Long	-0.10	-0.03	0.07	0.09	0.01	0.15	0.19	0.11	0.15
L-S	-0.25	-0.07	0.09	0.10	-0.03	0.25	0.27	0.14	0.22
t-stat	(-2.33)	(-1.10)	(1.49)	(1.33)	(-0.74)	(2.93)	(2.87)	(2.14)	(2.76)
Small Institutions									
Short	-0.47	-0.43	-0.41	-0.38	-0.42	-0.03	0.02	-0.12	-0.04
Long	-0.19	-0.28	-0.30	-0.34	-0.28	-0.06	-0.04	-0.08	-0.06
L-S	0.28	0.14	0.12	0.04	0.15	-0.03	-0.06	0.04	-0.02
t-stat	(4.70)	(4.25)	(2.19)	(0.79)	(3.55)	(-1.47)	(-2.01)	(2.14)	(-1.03)

Panel B: Institutional Trading based on ΔInst									
Long Horizon					Short Horizon				
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Large Institutions									
Short	15.59	15.69	13.42	11.77	14.12	0.65	-0.22	2.64	1.02
Long	9.59	10.65	12.14	14.31	11.67	7.60	9.86	6.58	8.12
L-S	-6.00	-5.03	-1.27	2.54	-2.44	6.94	10.09	3.91	7.09
t-stat	(-7.05)	(-10.30)	(-1.86)	(3.24)	(-5.43)	(20.25)	(11.62)	(9.37)	(15.33)
Medium Institutions									
Short	2.74	2.42	2.27	1.59	2.25	-0.93	-1.17	-0.29	-0.80
Long	0.50	1.54	1.65	2.56	1.56	2.07	2.44	1.67	2.06
L-S	-2.24	-0.89	-0.61	0.97	-0.69	3.00	3.61	1.96	2.85
t-stat	(-4.09)	(-4.62)	(-3.12)	(4.09)	(-4.15)	(11.38)	(13.32)	(7.97)	(11.23)
Small Institutions									
Short	1.30	1.14	1.10	0.76	1.07	-0.23	-0.26	0.04	-0.15
Long	0.47	0.62	0.79	1.13	0.75	0.80	0.91	0.67	0.79
L-S	-0.83	-0.51	-0.31	0.37	-0.32	1.03	1.17	0.63	0.94
t-stat	(-8.43)	(-8.24)	(-3.92)	(4.38)	(-6.25)	(13.18)	(17.12)	(8.52)	(13.23)

4.8 Tables 8–9: Liquidity preference heterogeneity

Table 8 shows that **large institutions** respond much more strongly to liquidity-driving events in their trading behavior than small institutions. Table 9 shows that this heterogeneity carries into anomaly trading:

- Large institutions drive the aggregate pattern: wrong-way trading on long-horizon anomalies and right-way trading on short-horizon anomalies.
- Smaller institutions show weaker or different patterns, consistent with weaker liquidity preference.

4.9 Table 10: Tick-size reforms as exogenous liquidity shocks

Table 10 uses tick-size reductions (1997 and 2001) as quasi-natural experiments.

Panel A. Liquidity improves (spreads fall), and institutional trading increases, especially for stocks experiencing large liquidity shocks.

Panel B. Within anomaly legs, the shock shifts both liquidity and institutional trading; importantly, the direction of institutional trading on long-horizon anomalies can change during the reform window.

Panel C (DiD). Using treated vs control (high-shock vs low-shock) comparisons within anomaly legs, the DiD estimates show that exogenous liquidity improvements causally raise institutional trading in the affected direction, supporting the paper’s causal mechanism.

4.10 Table 11: Asset pricing implications via institutional trading

Table 11 asks whether liquidity matters for anomaly *magnitudes* through institutional trading direction.

Table 10
Institutional trading on market anomalies: Exogenous liquidity shocks around the minimum tick size changes of 1997 and 2001.

Panel A: Effective Spread and Institutional Trading around Tick Size Changes								
	Effective Spread (%)				Δ inst (%)			
	Pre-Treatment Period	Treatment Period	Difference	t-Stat	Pre-Treatment Period	Treatment Period	Difference	t-Stat
1997 Event								
All Stocks	20.52	16.17	-4.34	(-23.78)	-0.28	4.85	5.13	(16.91)
Stocks with High Liquidity Shock	29.37	19.73	-9.63	(-33.55)	0.13	6.66	6.52	(13.71)
Stocks with Low Liquidity Shock	11.67	12.61	0.94	(8.04)	-0.69	3.04	3.73	(10.02)
High - Low	17.70	7.12	-10.58		0.82	3.62	2.80	
t-stat	(24.74)	(11.80)	(-34.09)		(2.13)	(9.28)	(4.63)	
2001 Event								
All Stocks	13.97	11.49	-2.41	(-18.86)	1.71	2.00	0.29	(1.07)
Stocks with High Liquidity Shock	17.28	12.15	-5.75	(-31.09)	0.48	3.34	2.86	(7.47)
Stocks with Low Liquidity Shock	10.65	10.83	0.93	(8.61)	2.93	0.66	-2.27	(-6.19)
High - Low	6.64	1.32	-6.67		-2.45	2.68	5.13	
t-stat	(13.37)	(2.80)	(-31.15)		(-6.15)	(8.59)	(9.68)	
Panel B: Institutional Trading on Anomalies around Tick Size Changes								
1997 Event: Long-Horizon Anomaly								
Short	19.72	15.46	-4.26	(-59.59)	0.92	4.15	3.23	(20.00)
Long	22.27	17.59	-4.69	(-63.44)	-1.21	5.17	6.38	(65.51)
Long-Short	2.55	2.13	-0.43		-2.13	1.02	3.14	
t-stat	(10.06)	(13.22)	(2.44)		(-34.00)	(13.48)	(30.25)	
1997 Event: Short-Horizon Anomaly								
Short	20.68	17.12	-3.56	(-36.75)	-0.33	3.14	3.47	(19.57)
Long	20.64	15.40	-5.24	(-48.68)	-0.10	6.41	6.51	(38.23)
Long-Short	-0.04	-1.72	-1.68		0.23	3.27	3.04	
t-stat	(-1.93)	(-11.55)	(-14.94)		(2.97)	(26.95)	(15.68)	
2001 Event: Long-Horizon Anomaly								
Short	13.64	11.48	-1.94	(-37.06)	2.97	1.81	-1.16	(-7.98)
Long	14.83	12.30	-2.74	(-55.18)	0.89	2.31	1.42	(16.35)
Long-Short	1.18	0.81	-0.80		-2.08	0.49	2.58	
t-stat	(8.88)	(6.21)	(-17.45)		(-27.37)	(3.29)	(22.23)	
2001 Event: Short-Horizon Anomaly								
Short	13.56	12.44	-1.34	(-19.80)	2.03	1.37	-0.66	(-3.23)
Long	14.10	10.58	-3.04	(-47.90)	1.85	2.82	0.97	(5.91)
Long-Short	0.54	-1.86	-1.70		-0.18	1.45	1.63	
t-stat	(2.07)	(-15.59)	(-23.20)		(-1.18)	(14.32)	(9.05)	
Panel C: DID Analysis: Institutional Trading on Anomalies around Tick Size Changes								
	Effective Spread (%)				Δ inst (%)			
	Pre-Treatment Period	Treatment Period	Difference	t-stat	Pre-Treatment Period	Treatment Period	Change in Δ inst From Previous Period	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1997 Event: Long-Horizon Anomaly								
Short Leg								
High Shock (Treated)	27.53	18.22	-9.31	(-80.35)	1.18	6.36	-3.58	5.18
Low Shock (Control)	12.03	12.75	0.72	(15.47)	0.65	1.98	-4.10	1.33
Treated-Control	15.50	5.47	-10.04		0.53	4.38	0.52	3.85
t-stat	(5.67)	(2.41)	(-8.09)		(2.21)	(23.99)	(1.76)	(13.41)
Long Leg								
High Shock (Treated)	31.87	21.51	-10.37	(-89.77)	-1.63	6.95	-4.19	8.58
Low Shock (Control)	12.78	13.72	0.94	(21.33)	-0.8	3.40	-4.12	4.20
Treated-Control	19.09	7.79	-11.30		-0.83	3.55	-0.07	4.38
t-stat	(8.46)	(3.15)	(-9.12)		(-6.45)	(23.52)	(-0.32)	(19.89)
1997 Event: Short-Horizon Anomaly								
Short Leg								
High Shock (Treated)	28.07	19.43	-8.64	(-57.95)	-0.18	4.72	-4.19	4.90
Low Shock (Control)	13.37	14.84	1.47	(19.92)	-0.48	1.58	-4.15	2.06
Treated-Control	14.70	4.60	-10.10		0.30	3.14	-0.04	2.84
t-stat	(3.75)	(1.39)	(-6.09)		(1.16)	(13.48)	(-0.07)	(7.98)
Long Leg								
High Shock (Treated)	30.19	19.40	-10.79	(-62.25)	-0.26	8.43	-3.73	8.69
Low Shock (Control)	11.16	11.43	0.27	(4.74)	0.06	4.41	-3.93	4.36
Treated-Control	19.03	7.98	-11.06		-0.32	4.02	0.20	4.33
t-stat	(4.83)	(2.54)	(-6.04)		(-1.27)	(18.30)	(0.62)	(12.72)

Table 10 (continued).

2001 Event: Long-Horizon Anomaly								
Short Leg								
High Shock (Treated)	16.17	11.32	-5.28	(-67.89)	1.35	3.19	0.55	1.83
Low Shock (Control)	11.17	11.64	1.33	(20.98)	4.55	0.47	0.41	-4.09
Treated-Control	5.00	-0.32	-6.60		-3.20	2.72	0.14	5.92
t-stat	(2.63)	(-0.18)	(-7.44)		(-14.49)	(19.27)	(1.30)	(21.54)
Long Leg								
High Shock (Treated)	18.54	13.26	-6.27	(-85.52)	0.12	3.51	0.03	3.38
Low Shock (Control)	11.17	11.35	0.74	(15.97)	1.65	1.12	0.15	-0.52
Treated-Control	7.37	1.91	-7.01		-1.53	2.39	-0.12	3.90
t-stat	(3.74)	(1.06)	(-8.20)		(-12.85)	(20.39)	(-0.95)	(22.12)
2001 Event: Short-Horizon Anomaly								
Short Leg								
High Shock (Treated)	15.01	11.40	-4.63	(-47.80)	0.69	2.42	-0.23	1.73
Low Shock (Control)	12.15	13.46	1.90	(26.12)	3.24	0.33	-0.33	-3.01
Treated-Control	2.86	-2.06	-6.53		-2.65	2.09	0.10	4.74
t-stat	(1.18)	(-0.69)	(-5.41)		(-10.96)	(12.59)	(0.27)	(15.23)
Long Leg								
High Shock (Treated)	18.48	12.16	-6.28	(-64.80)	0.74	4.48	0.86	3.74
Low Shock (Control)	9.78	9.02	0.16	(3.31)	2.94	1.18	0.89	-1.76
Treated-Control	8.71	3.13	-6.43		-2.20	3.30	-0.03	5.50
t-stat	(3.45)	(1.42)	(-6.06)		(-9.56)	(17.99)	(-0.16)	(17.89)

This table reports changes (in percentage points, %) in liquidity and institutional trading around the events of minimum tick size change in 1997 and 2001. The treatment periods are the year 1997 and the first half of 2001, with the corresponding pre-treatment periods being the year 1996 and the first half of 2000. Technology stocks are excluded from the analysis for the 2001 event. Liquidity is measured by the value-weighted percentage effective spread. Institutional trading, Δ inst, is the change in the fraction of shares held by institutions. Stocks with high (low) liquidity shocks are those with above-median (below-median) reduction in the effective spread from the pre-treatment to the treatment period. Panel A presents changes (differences) in the effective spread and institutional trading from the pre-treatment to the treatment periods, averaged across all stocks, stocks with high liquidity shocks, or stocks with low liquidity shocks. Panel B reports changes in the effective spread and institutional trading for the long and short legs of anomaly portfolios from the pre-treatment to the treatment periods. Panel C presents changes in the effective spread and institutional trading for stocks with high and low liquidity shocks in the long and short legs of anomaly portfolios from the pre-treatment to the treatment periods. *t*-statistics are reported in parentheses.

Table 11
Magnitude of market anomalies conditional on institutional trading.

Portfolios	Long Horizon					Short Horizon			
	Value	Inv. & Fin.	Intangible	LT Prof.	LH Avg.	Earnings Mom.	Return Mom.	ST Prof.	SH Avg.
Panel A. Conditional on the Direction of Institutional Trading $\Delta Inst$									
LH	-0.03	-0.04	-0.03	0.12	0.00	0.34	0.42	0.57	0.44
t-stat	(-0.17)	(-0.20)	(-0.10)	(0.47)	(0.02)	(1.24)	(1.27)	(1.90)	(1.53)
SL	-0.70	-0.53	-0.48	-1.02	-0.68	-1.11	-1.25	-1.56	-1.31
t-stat	(-3.20)	(-2.48)	(-2.27)	(-5.35)	(-3.72)	(-4.44)	(-4.39)	(-5.90)	(-5.18)
LL	0.68	0.43	0.60	1.04	0.69	0.19	0.38	0.55	0.37
t-stat	(2.96)	(2.51)	(2.42)	(4.29)	(3.42)	(0.86)	(1.67)	(2.33)	(1.81)
SH	-1.01	-1.20	-1.16	-1.31	-1.17	-0.90	-1.49	-1.17	-1.19
t-stat	(-4.16)	(-5.39)	(-5.15)	(-4.45)	(-5.18)	(-4.06)	(-5.18)	(-3.60)	(-4.93)
LH-SL (Right)	0.66	0.49	0.45	1.14	0.69	1.44	1.67	2.13	1.75
t-stat	(2.54)	(1.65)	(1.15)	(3.60)	(2.65)	(3.47)	(3.26)	(4.52)	(3.95)
LL-SH (Wrong)	1.69	1.63	1.76	2.34	1.86	1.08	1.88	1.72	1.56
t-stat	(4.66)	(6.00)	(5.08)	(5.59)	(5.99)	(3.58)	(4.49)	(3.64)	(4.64)
Wrong-Right	1.03	1.14	1.31	1.20	1.17	-0.36	0.21	-0.42	-0.19
t-stat	(2.77)	(2.55)	(2.88)	(2.58)	(2.74)	(-0.70)	(0.53)	(-0.78)	(-0.41)
Panel B. Conditional on the Direction of Liquidity-Related Institutional Trading $\Delta Inst^{LIQ}$									
LH	-0.13	-0.51	-0.35	-0.09	-0.27	0.28	0.25	0.59	0.37
t-stat	(-0.61)	(-2.37)	(-1.29)	(-0.38)	(-1.38)	(0.96)	(0.72)	(1.87)	(1.22)
SL	-0.32	-0.12	-0.04	-0.35	-0.21	-0.94	-1.14	-1.24	-1.11
t-stat	(-1.13)	(-0.45)	(-0.17)	(-1.50)	(-0.87)	(-3.99)	(-4.26)	(-4.78)	(-4.52)
LL	0.84	0.58	0.80	1.12	0.83	0.60	0.81	1.07	0.83
t-stat	(3.30)	(2.69)	(3.17)	(4.08)	(3.65)	(2.59)	(3.45)	(3.89)	(3.67)
SH	-1.64	-1.64	-1.57	-2.06	-1.73	-1.23	-1.56	-1.65	-1.48
t-stat	(-7.05)	(-7.35)	(-6.02)	(-6.29)	(-7.17)	(-4.71)	(-6.74)	(-6.69)	(-6.22)
LH-SL (Right)	0.19	-0.38	-0.31	0.26	-0.06	1.22	1.38	1.83	1.48
t-stat	(0.51)	(-0.94)	(-0.73)	(0.75)	(-0.18)	(2.78)	(2.63)	(3.68)	(3.13)
LL-SH (Wrong)	2.48	2.22	2.37	3.17	2.56	1.83	2.36	2.71	2.30
t-stat	(6.35)	(6.48)	(5.75)	(6.09)	(6.69)	(4.50)	(6.53)	(5.22)	(6.27)
Wrong-Right	2.29	2.60	2.68	2.92	2.62	0.61	0.98	0.88	0.82
t-stat	(3.82)	(3.78)	(3.98)	(4.12)	(3.99)	(0.87)	(1.77)	(1.24)	(1.28)
Panel C. Conditional on the Direction of Non-Liquidity Institutional Trading $\Delta Inst^{NLQ}$									
LH	0.10	0.07	0.15	0.27	0.15	0.40	0.54	0.65	0.53
t-stat	(0.42)	(0.33)	(0.55)	(1.09)	(0.68)	(1.56)	(1.81)	(2.41)	(2.03)
SL	-1.34	-1.12	-1.08	-1.66	-1.30	-1.27	-1.46	-1.87	-1.54
t-stat	(-6.62)	(-6.18)	(-5.38)	(-7.21)	(-7.34)	(-5.47)	(-5.50)	(-6.84)	(-6.54)
LL	0.42	0.01	0.25	0.57	0.31	0.11	0.24	0.48	0.28
t-stat	(2.07)	(0.08)	(1.15)	(2.79)	(1.94)	(0.50)	(0.96)	(2.03)	(1.30)
SH	-0.84	-0.97	-0.87	-1.03	-0.93	-0.90	-1.42	-1.14	-1.15
t-stat	(-3.46)	(-4.38)	(-4.17)	(-3.67)	(-4.21)	(-4.26)	(-4.82)	(-3.59)	(-4.66)
LH-SL (Right)	1.44	1.18	1.24	1.92	1.45	1.68	2.00	2.52	2.07
t-stat	(4.91)	(4.84)	(3.47)	(5.28)	(5.49)	(4.58)	(4.31)	(5.67)	(5.30)
LL-SH (Wrong)	1.26	0.98	1.13	1.60	1.24	1.01	1.65	1.62	1.43
t-stat	(3.80)	(4.27)	(3.75)	(4.78)	(5.00)	(3.21)	(3.63)	(3.50)	(4.00)
Wrong-Right	-0.18	-0.20	-0.11	-0.32	-0.20	-0.67	-0.35	-0.90	-0.64
t-stat	(-0.52)	(-0.57)	(-0.31)	(-0.84)	(-0.58)	(-1.53)	(-0.90)	(-2.02)	(-1.55)

Panel A (overall $\Delta Inst$). For long-horizon anomalies, FF3 alphas are larger when institutions trade in the *wrong* direction (LL-SH) than when they trade in the *right* direction (LH-SL). This suggests wrong-way trading *exacerbates* these anomalies.

Panel B (liquidity-related $\Delta Inst^{LIQ}$). The wrong-minus-right difference becomes even larger, indicating that the *liquidity-driven* component of institutional trading is the key channel. **Panel C (non-liquidity $\Delta Inst^{NLQ}$).** The wrong-minus-right difference is not positive for long-horizon anomalies, consistent with the idea that *liquidity, not other trading motives*, is what drives the amplification.

5 Unified Mechanism: One Story Linking All Results

The paper's results can be read as a coherent chain:

1. Anomalies differ in return horizons (Table 1), implying different underlying mispricing dynamics (convergence vs divergence).
2. The anomaly legs systematically differ in liquidity levels and liquidity changes (Table 2, Fig. 1).
3. Institutions trade along the liquidity gradients (Table 3, Fig. 1), and the liquidity-related trading component explains most of the pattern (Table 4).
4. The liquidity gradients are traced to plausibly exogenous liquidity-driving factors (Tables 5–6, Fig. 2), and those factors generate an event-driven component of institutional trading that matches the anomaly-trading patterns (Table 7).

5. Liquidity preference explains why large institutions dominate the effect (Tables 8–9).
6. Quasi-natural experiments (tick-size reforms) provide causal evidence that liquidity shocks move institutional trading in the predicted way (Table 10).
7. Liquidity matters for market efficiency mainly by shaping institutional trading direction, which affects anomaly magnitudes (Table 11), not primarily through a simple “liquidity premium” explanation.

6 Conclusion

6.1 Summary

This paper concludes that **market anomalies are systematically linked to liquidity in a horizon-dependent way**, and that this connection helps explain **why institutional investors sometimes trade “the wrong way” on anomalies** and why **liquidity can worsen (not necessarily fix) mispricing**.

The key chain of logic is:

1. **Anomalies are not liquidity-neutral.** Long-short anomaly portfolios have distinct *liquidity characteristics* that vary with the anomaly’s *return-predictive horizon*. Using the Daniel et al. (2020) classification, the paper studies seven anomaly categories, where *long-horizon* categories (e.g., value; investment and financing; intangible investments; long-term profitability) predict returns beyond one year, while *short-horizon* categories (e.g., earnings momentum; return momentum; short-term profitability) predict returns within one year.
2. **Long-horizon anomalies resemble liquidity provision; short-horizon anomalies resemble liquidity demand.** For *long-horizon* anomalies, stocks in the *long legs* tend to be **less liquid** and exhibit **smaller liquidity improvements** than stocks in the short legs. Consequently, long-horizon long-short strategies have a **liquidity-provision** feature (analogous to on/off-the-run convergence trades). For *short-horizon* anomalies, the pattern reverses: stocks in the long legs have **better liquidity** and **greater liquidity improvements** than stocks in the short legs, so these strategies are **liquidity-demanding**.
3. **Institutional trading direction lines up with liquidity, not with anomaly signals.** Net institutional trading—defined as the difference in institutional ownership changes between the long and short legs—is **negative (wrong-direction)** prior to formation for long-horizon anomalies and **positive (right-direction)** for short-horizon anomalies. This pattern is largely explained by a *liquidity-related component* of institutional trading.
4. **Exogenous liquidity drivers and institutional liquidity preferences play causal roles.** Liquidity characteristics can be traced to liquidity-driving events (financing events, mergers, dividend changes, stock splits, index membership changes), as well as changes in analyst coverage and news coverage. An “event-driven” institutional trading component built from institutions’ responses to these liquidity drivers explains **the bulk** (about **85%**) of wrong-direction net

trading on long-horizon anomalies. Additional quasi-natural experiments—minimum tick size changes in 1997 and 2001—generate exogenous liquidity shocks that move institutional trading in the predicted direction, and the response is stronger for institutions with stronger liquidity preferences (proxied by portfolio size).

5. **Liquidity can exacerbate anomaly magnitudes.** For long-horizon anomalies, anomaly returns are **larger** when liquidity-related institutional trading is in the wrong direction than when it is in the right direction. By contrast, the component of institutional trading *unrelated* to liquidity has little effect on anomaly magnitude. Hence liquidity does not always enhance market efficiency; it can **amplify** anomalies through liquidity-motivated institutional trading.

6.2 One unified mechanism

The intuition can be summarized by three interacting ingredients: (i) *horizon-dependent liquidity asymmetry across the anomaly legs*, (ii) *institutional liquidity preference*, and (iii) *liquidity shocks*.

1) Horizon-dependent liquidity asymmetry across the long and short legs. Long-horizon anomaly signals tend to be tied to slow-moving firm changes (financing and investment growth, intangible investment accumulation, long-run profitability) that are also associated with *liquidity-improving* corporate events and attention. These liquidity improvements disproportionately appear in stocks that end up in the *short legs* of long-horizon anomalies, so the long leg stays relatively illiquid while the short leg becomes more liquid. Short-horizon anomalies are tied to recent information and near-term profitability/earnings news, which is often accompanied by rapid trading and attention, making the *long legs* of these anomalies relatively more liquid and faster-improving.

2) Institutions have a preference for liquidity. Large institutions face trading-cost and capacity constraints. They therefore tilt toward assets whose liquidity is high or improving. This preference implies they tend to buy the more liquid side of a long–short portfolio.

3) Putting (1) and (2) together explains the “wrong-way” vs “right-way” trading.

- For **long-horizon anomalies**, the *short leg* is typically more liquid / improving. Institutions then buy relatively more of the short-leg stocks and sell relatively more of the long-leg stocks, producing **negative net institutional trading**—the wrong direction relative to the anomaly signal.
- For **short-horizon anomalies**, the *long leg* is typically more liquid / improving. Institutions then buy more of the long leg and sell more of the short leg, producing **positive net institutional trading**—the right direction.

4) Why liquidity can *amplify* anomalies. When institutions trade primarily to satisfy liquidity preference rather than to correct mispricing, liquidity improvements can steer demand toward the “liquid” leg even if that leg is *overvalued* (the short leg in long-horizon anomalies). This demand pressure can move prices further away from fundamentals, increasing the long–short return spread.

Thus, unlike the classic “illiquidity limits arbitrage” narrative, the paper highlights a distinct channel:

Liquidity shocks $\Rightarrow$ liquidity-motivated institutional trading $\Rightarrow$ larger anomaly magnitudes.

6.3 Takeaway

Cao, Liang, Yao, and Zhang (2026) show that anomalies have systematic, horizon-dependent liquidity characteristics that shape institutional trading: institutions chase liquidity (buy the more liquid leg), which makes them trade against long-horizon anomalies but in favor of short-horizon anomalies, and exogenous liquidity shocks plus institutional liquidity preference can causally drive these patterns and even amplify anomaly returns.